

Lizhong Hu

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Profile

Software Engineering undergraduate in a Sino-American dual-degree program, with experience in back-end development, RAG systems, Linux/Docker infrastructure, and live sports broadcast support.

Research Interests

Algorithms and Data Structures; Software Systems; Information Retrieval and Retrieval-Augmented Generation.

Education

Quanzhou University of Information Engineering

2023–2027 (*expected*)

Software Engineering (Sino-foreign Cooperation)

GPA: 3.10/4.00

Slippery Rock University

Information Systems, Bachelor of Science (BS)

GPA: 3.583/4.000

Sino-American 4+0 dual-degree program conducted in China; approved by the Ministry of Education of China.

Selected Projects

NoteLLM: Personal Learning QA System with Verifiable RAG

Ongoing

Undergraduate thesis

- Built a RAG backend using FastAPI, PostgreSQL, and `pgvector` for cosine-similarity retrieval, with page-aware extraction and overlapping chunking across PDF, Markdown, and TXT files.
- Validated citations against retrieved chunks and stored source excerpts. A single baseline on 7 synthetic documents and 34 questions recorded 100% source-level Recall@5 and 97.1% automatic citation-to-source match.

DOMjudge Online Judge Deployment & Operations

Sep 2024–Present

- Maintain containerized DOMjudge for campus training and programming contests.

Encrypted Overlay Network with Dual-Stack Egress Isolation

- Managed network configurations using Cloudflare Workers and Linux VPS, with proxy profile distribution and IPv4/IPv6 routing; added node selection, repeatable deployment, backups, connectivity checks, and regression tests.

ETIPlus: Community Governance & WeChat Group Management Backend

Mar 2025–Apr 2025

- Built a Flask backend with PostgreSQL for landlord credit scoring and anomaly audits, and MongoDB for WeChat message storage.
- Developed APIs for group moderation, regex nickname checks, and batch imports; automated regression checks with Postman.

Technical Experience

Sinotech Xinrui (Beijing) Co., Ltd.

7–22 Sep 2026

TOC Operator / Gifu, Japan

Project: 20th Asian Games Aichi-Nagoya 2026 — broadcast technical support.

Football, Gifu Nagaragawa Stadium.

Host broadcaster: China Media Group-International Media Port (CMG-IMP).

- Checked power, intercom quality, and primary/backup feeds with the broadcast team before matches.
- Monitored TOC signals during matches and provided on-site technical support throughout a 16-day assignment.
- Helped engineers isolate an optical-transmitter fault through receiver replacement and attenuation tests; the monitor feed was restored after post-match replacement and end-to-end verification.

NOI-Pre Problem Bank — Matiji

Jul 2024–Oct 2024

Problem Tester

In collaboration with China Computer Federation (CCF).

- Tested NOI-Pre programming problems; created test cases, checked solution correctness through targeted submissions, and reported issues with test data and judge results.

Competitive Programming & Awards

- 2026 National Invitational of CCPC (Fujian), The 13th Fujian Collegiate Programming Contest** — First accepted solution to Problem A.
- The 2024 ICPC Asia Kunming Regional Contest** — Honorable Medal, 2024.
- 15th Lanqiao Cup** — First Prize, Fujian Provincial Round, Python Programming (University Group B), 2024.

Technical Skills

Languages & Algorithms: C++, Python, Java, JavaScript, Go; algorithms and data structures.

Back-End & Data: FastAPI, Flask, RuoYi, PostgreSQL, MongoDB, MySQL, REST API design and integration.

Systems & DevOps: Linux, Docker, Git, DOMjudge, Cloudflare Workers/CDN/DNS, WARP routing.

Information Retrieval & AI: Retrieval-Augmented Generation (RAG), LLM API integration, prompt design.

AI Coding Tools: Familiar with Codex, Claude Code, and Antigravity CLI; repository instructions, shared Skills, parallel subagent review, and test/build validation.

Contest Authoring Tools: Familiar with Polygon (Codeforces) and Nowcoder; problem preparation, test data, and solution verification.